Chapter 5

Cultivation of Microorganisms

ENI9S6

Chapter Outline

5.1 Significance of Culturing Microorganisms

5.2 Bacteriological Media and its Types

5.3 Pure Culture

5.4 Growth and Colony Characteristics of Bacteria and Fungi



The microorganisms on the handprint of an eight-year-old boy. After incubation the plates showed coloured colonies of bacteria and fungi as you see above.

The cultivation of microorganisms under laboratory condition makes the microscopic cells to grow and form individual colonies macroscopically.

Learning Objectives

After studying this chapter the student will be able,

- *To know the importance of bacterial media for growth of microorganisms. To understand various types of media for differentiation and diagnosis of important pathogenic microorganisms.*
- *To know pure culture techniques*
- *To understand the methods involved in isolating pure culture of bacteria, which includes pour plate, spread plate and streak plate.*
- *To differentiate the growth characteristics of bacteria and fungi.*

Microorganisms are omnipresent and they exist in soil, air, water, spoiled food, decayed animal and plant residues. They are found in environment as pathogens and normal microflora. Excellent supporting factors are available in nature for microorganisms to survive in the environment. This leads to microbial proliferation as an extended community in nature. The term 'cultivation of microorganisms' means growing microorganisms in the laboratory with ample supply of specific nutrients (Figure 5.1). Obligate intracellular parasites like viruses, *Rickettsias* and *Chlamydias* are cultivated within living cells.

Survival and growth of microorganisms depend upon the favourable growth environment. Laboratory cultivation plays a crucial role in the isolation, identification and classification of microorganisms. Cultivation of bacteria and fungi by artificial formulated medium is one of the important milestones in the history of Microbiology.